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CS-158A

Lab 3

Checkpoint 1

Wireshark packet capture showing a SYN packet from 128.119.245.12 to 10.250.47.218. The packet has SYN=1 and ISN=1308706865. The packet details pane shows the TCP header with SYN=1 and ISN=1308706865.

SYN packet has SYN bit set only and ISN 1308706865

Wireshark packet capture showing a SYN-ACK packet from 10.250.47.218 to 128.119.245.12. The packet has SYN=1 and ACK=1. The packet details pane shows the TCP header with SYN=1 and ACK=1.

SYN-ACK has both SYN and ACK bits set and ISN 878333915

Wireshark packet capture analysis of a TCP connection. The packet list shows a SYN packet (No. 533) and its corresponding ACK (No. 537). The packet details for the ACK packet (No. 537) are expanded, showing the following fields:

- Frame 533: Packet, 1440 bytes on wire (11520 bits), 1440 bytes captured (11520 bits) on interface wlp1s0, id 0
- Ethernet II, Src: de:8e:d6:a6:63:8c (de:8e:d6:a6:63:8c), Dst: Cisco:39:70:1f (60:26:aa:39:70:1f)
- Internet Protocol Version 4, Src: 10.250.47.218, Dst: 128.119.245.12
- Transmission Control Protocol, Src Port: 57792, Dst Port: 80, Seq: 1, Len: 1374
- Source Port: 57792
- Destination Port: 80
- [Stream index: 5]
- [Stream Packet Number: 4]
- [Conversation completeness: Complete, WITH_DATA (31)]
- [TCP Segment Len: 1374]
- Sequence Number: 1 (relative sequence number)
- Sequence Number (raw): 1308706866
- Next sequence number: 1375 (relative sequence number)
- Acknowledgment Number: 1 (relative ack number)
- Acknowledgment number (raw): 878333916
- 1000 = Header Length: 32 bytes (8)
- Flags: 0x010 (ACK)
- 0000 = Reserved: Not set
- 0000 = Accurate ECN: Not set
- 0000 = Congestion Window Reduced: Not set
- 0000 = ECN-Echo: Not set
- 0000 = Urgent: Not set
- 0000 = Acknowledgment: Set
- 0000 = Push: Not set
- 0000 = Reset: Not set
- 0000 = SYN: Not set
- 0000 = FIN: Not set
- [TCP Flags:A....]
- Window: 63
- [Calculated window size: 64512]
- [Window size scaling factor: 1024]
- Checksum status: Unverified
- [Checksum Status: Unverified]
- Next Sequence Number (tcp.nextseq)

The packet bytes pane shows the raw data of the ACK packet, including the Ethernet II header, IP header, and TCP header. The packet is identified as a SYN-ACK packet (Seq=1, Ack=1) from 10.250.47.218 to 128.119.245.12.

Source port 57792, destination port 80, ISN 1308706866, ACK 878333916, ACK flag set, Window size 64512

Syn to Syn-Ack time 0.074256292 s

Checkpoint 2

Segment	Seq (relative)	Length	Predict ACK	Actual ACK
533	1	1374	1374	2749
534	1375	1375	2749	2749
535	2749	1374	4123	5497

capture.pcapng

ip.addr == 128.119.245.12 && (tcp.flags.syn == 1 || tcp.flags.ack == 1)

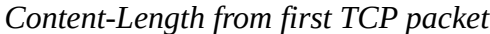
No.	Time	Source	Destination	Protocol	Length	Info
533	1.087933488	128.119.245.12	128.119.245.12	TCP	68	68 -- 57792 -- 80 [SYN] Seq=0 Win=64248 Len=0 MSS=1460 SACK_PERM TSval=136491986 TSecr=0 WS=1024
534	1.162189780	128.119.245.12	128.119.245.12	TCP	74	80 -- 57792 [SYN, ACK] Seq=0 Ack=1 Win=85108 Len=0 MSS=1386 SACK_PERM TSval=3804060893 TSecr=136491006 WS=128
535	1.162279207	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 -- 80 [ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval=136492061 TSecr=3804060893
605	1.162561023	128.119.245.12	128.119.245.12	TCP	1440	57792 -- 80 [ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval=136492061 TSecr=3804060893 [TCP PDU reassembled in 776]
606	1.162561823	128.119.245.12	128.119.245.12	TCP	1440	57792 -- 80 [PSH, ACK] Seq=13745 Ack=1 Win=64512 Len=1374 TSval=136492061 TSecr=3804060893 [TCP PDU reassembled in 776]
607	1.162688805	128.119.245.12	128.119.245.12	TCP	1440	57792 -- 80 [ACK] Seq=2749 Ack=1 Win=64512 Len=1374 TSval=136492061 TSecr=3804060893 [TCP PDU reassembled in 776]
608	1.162688805	128.119.245.12	128.119.245.12	TCP	1440	57792 -- 80 [PSH, ACK] Seq=4123 Ack=1 Win=64512 Len=1374 TSval=136492061 TSecr=3804060893 [TCP PDU reassembled in 776]
609	1.162688805	128.119.245.12	128.119.245.12	TCP	1440	57792 -- 80 [ACK] Seq=5497 Ack=1 Win=64512 Len=1374 TSval=136492062 TSecr=3804060893 [TCP PDU reassembled in 776]
610	1.163631786	128.119.245.12	128.119.245.12	TCP	1440	57792 -- 80 [PSH, ACK] Seq=6071 Ack=1 Win=64512 Len=1374 TSval=136492062 TSecr=3804060893 [TCP PDU reassembled in 776]
611	1.164260990	128.119.245.12	128.119.245.12	TCP	1440	57792 -- 80 [ACK] Seq=8245 Ack=1 Win=64512 Len=0 TSval=136492063 TSecr=3804060893 [TCP PDU reassembled in 776]
612	1.164260937	128.119.245.12	128.119.245.12	TCP	1440	57792 -- 80 [PSH, ACK] Seq=9639 Ack=1 Win=64512 Len=1374 TSval=136492063 TSecr=3804060893 [TCP PDU reassembled in 776]
613	1.164952539	128.119.245.12	128.119.245.12	TCP	1440	57792 -- 80 [ACK] Seq=10993 Ack=1 Win=64512 Len=1374 TSval=136492063 TSecr=3804060893 [TCP PDU reassembled in 776]
614	1.164954633	128.119.245.12	128.119.245.12	TCP	1440	57792 -- 80 [PSH, ACK] Seq=12367 Ack=1 Win=64512 Len=1374 TSval=136492063 TSecr=3804060893 [TCP PDU reassembled in 776]
615	1.249329191	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 [ACK] Seq=1 Ack=13745 Win=66832 Len=0 TSval=136492061 TSecr=136492061
616	1.249329191	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 [ACK] Seq=1 Ack=13745 Win=66832 Len=0 TSval=136492061 TSecr=136492061
617	1.249329191	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 [ACK] Seq=1 Ack=13745 Win=66832 Len=0 TSval=136492061 TSecr=136492061
618	1.249329191	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 [ACK] Seq=1 Ack=13745 Win=66832 Len=0 TSval=136492061 TSecr=136492061
619	1.249329191	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 [ACK] Seq=1 Ack=13745 Win=66832 Len=0 TSval=136492061 TSecr=136492061
620	1.249329191	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 [ACK] Seq=1 Ack=13745 Win=66832 Len=0 TSval=136492061 TSecr=136492061
621	1.249329191	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 [ACK] Seq=1 Ack=13745 Win=66832 Len=0 TSval=136492061 TSecr=136492061
622	1.249329191	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 [ACK] Seq=1 Ack=13745 Win=66832 Len=0 TSval=136492061 TSecr=136492061
623	1.249329191	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 [ACK] Seq=1 Ack=13745 Win=66832 Len=0 TSval=136492061 TSecr=136492061
624	1.249329191	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 [ACK] Seq=1 Ack=13745 Win=66832 Len=0 TSval=136492061 TSecr=136492061
625	1.249329191	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 [ACK] Seq=1 Ack=13745 Win=66832 Len=0 TSval=136492061 TSecr=136492061
626	1.249329191	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 [ACK] Seq=1 Ack=13745 Win=66832 Len=0 TSval=136492061 TSecr=136492061
627	1.249329191	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 [ACK] Seq=1 Ack=13745 Win=66832 Len=0 TSval=136492061 TSecr=136492061
628	1.249329191	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 [ACK] Seq=1 Ack=13745 Win=66832 Len=0 TSval=136492061 TSecr=136492061
629	1.249329191	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 [ACK] Seq=1 Ack=13745 Win=66832 Len=0 TSval=136492061 TSecr=136492061
630	1.249329191	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 [ACK] Seq=1 Ack=13745 Win=66832 Len=0 TSval=136492061 TSecr=136492061
631	1.249329191	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 [ACK] Seq=1 Ack=13745 Win=66832 Len=0 TSval=136492061 TSecr=136492061
632	1.249329191	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 [ACK] Seq=1 Ack=13745 Win=66832 Len=0 TSval=136492061 TSecr=136492061

Packets 533-535 are sent from client. 605 ACKs 533-534, 606 ACKs 535

ip.addr == 128.119.245.12 && (tcp.flags.syn == 1 || tcp.flags.ack == 1)

No.	Time	Source	Destination	Protocol	Length	Info
763	1.417611448	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 [ACK] Seq=1 Ack=90655 Win=122112 Len=0 TSval=3804060346 TSecr=136492226
820	1.417615536	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 [ACK] Seq=1 Ack=92059 Win=124800 Len=0 TSval=3804060401 TSecr=136492296
821	1.417646289	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 [ACK] Seq=1 Ack=93433 Win=127616 Len=0 TSval=3804060402 TSecr=136492296
822	1.417646930	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 [ACK] Seq=1 Ack=94897 Win=130984 Len=0 TSval=3804060402 TSecr=136492296
823	1.417646930	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 [ACK] Seq=1 Ack=95555 Win=133680 Len=0 TSval=3804060402 TSecr=136492296
824	1.417647211	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 [ACK] Seq=1 Ack=96929 Win=136824 Len=0 TSval=3804060402 TSecr=136492296
825	1.417647341	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 [ACK] Seq=1 Ack=100363 Win=141312 Len=0 TSval=3804060402 TSecr=136492296
826	1.417647471	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 [ACK] Seq=1 Ack=104425 Win=145064 Len=0 TSval=3804060403 TSecr=136492296
827	1.417647611	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 [ACK] Seq=1 Ack=107173 Win=150000 Len=0 TSval=3804060403 TSecr=136492296
828	1.417647742	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 [ACK] Seq=1 Ack=110125 Win=153528 Len=0 TSval=3804060403 TSecr=136492296
829	1.417647889	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 [ACK] Seq=1 Ack=114843 Win=158832 Len=0 TSval=3804060403 TSecr=136492296
830	1.417648029	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 [ACK] Seq=1 Ack=119517 Win=164544 Len=0 TSval=3804060403 TSecr=136492296
831	1.417648310	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 [ACK] Seq=1 Ack=124791 Win=170336 Len=0 TSval=3804060403 TSecr=136492296
832	1.417648510	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 [ACK] Seq=1 Ack=129539 Win=176184 Len=0 TSval=3804060403 TSecr=136492296
833	1.417648640	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 [ACK] Seq=1 Ack=134287 Win=182016 Len=0 TSval=3804060403 TSecr=136492296
834	1.417648818	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 [ACK] Seq=1 Ack=139061 Win=187872 Len=0 TSval=3804060404 TSecr=136492296
835	1.417649214	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 [ACK] Seq=1 Ack=143849 Win=193728 Len=0 TSval=3804060404 TSecr=136492296
836	1.417649299	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 [ACK] Seq=1 Ack=148619 Win=199648 Len=0 TSval=3804060404 TSecr=136492296
837	1.417649439	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 [ACK] Seq=1 Ack=153395 Win=205504 Len=0 TSval=3804060404 TSecr=136492296
838	1.417649569	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 [ACK] Seq=1 Ack=158169 Win=211392 Len=0 TSval=3804060404 TSecr=136492296
839	1.417649700	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 [ACK] Seq=1 Ack=162943 Win=217280 Len=0 TSval=3804060404 TSecr=136492296
840	1.417649830	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 [ACK] Seq=1 Ack=167717 Win=223168 Len=0 TSval=3804060405 TSecr=136492296
841	1.417649970	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 [ACK] Seq=1 Ack=172491 Win=229056 Len=0 TSval=3804060405 TSecr=136492296
842	1.417650109	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 [ACK] Seq=1 Ack=177265 Win=234944 Len=0 TSval=3804060405 TSecr=136492296
843	1.417650249	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 [ACK] Seq=1 Ack=182039 Win=240832 Len=0 TSval=3804060405 TSecr=136492296
844	1.417650389	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 [ACK] Seq=1 Ack=186813 Win=246720 Len=0 TSval=3804060405 TSecr=136492296
845	1.417650529	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 [ACK] Seq=1 Ack=191587 Win=252608 Len=0 TSval=3804060405 TSecr=136492296
846	1.417650669	128.119.245.12	128.119.245.12	TCP	66	80 -- 57792 [ACK] Seq=1 Ack=196361 Win=258496 Len=0 TSval=3804060405 TSecr=136492296
847	1.417650809	128.119.245.12	128.119.245.12	TCP	66	57792 -- 80 [ACK] Seq=125953 Ack=781 Win=64512 Len=0 TSval=136492276 TSecr=3804060407
890	1.508760911	128.119.245.12	128.119.245.12	HTTP	454	GET /favicon.ico HTTP/1.1
891	1.635354852	128.119.245.12	128.119.245.12	HTTP	648	HTTP/1.1 301 Moved Permanently (text/html)
892	1.635363932	128.119.245.12	128.119.245.12	TCP	66	57792 -- 80 [ACK] Seq=155341 Ack=1363 Win=64512 Len=0 TSval=136492532 TSecr=3804060564
920	1.64410613	128.119.245.12	128.119.245.12	TCP	74	40598 -- 443 [SYN] Seq=0 Win=64248 Len=0 MSS=1460 SACK_PERM TSval=3804060562 TSecr=0 WS=1024
929	1.725022298	128.119.245.12	128.119.245.12	TCP	74	443 -- 40598 [SYN, ACK] Seq=0 Ack=1 Win=64512 Len=0 MSS=1386 SACK_PERM TSval=3804060562 TSecr=3823488392 WS=128
929	1.725153957	128.119.245.12	128.119.245.12	TCP	66	40598 -- 443 [ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval=3823488473 TSecr=3804060562
930	1.728071387	128.119.245.12	128.119.245.12	TCP	1440	40598 -- 443 [ACK] Seq=1 Ack=1 Win=64512 Len=1374 TSval=13649288476 TSecr=3804060562 [TCP PDU reassembled in 931]
931	1.728079762	128.119.245.12	128.119.245.12	TLSv1.3	1268	Client Hello (SN[gaia.cs.umass.edu])
944	1.800784394	128.119.245.12	128.119.245.12	TCP	66	443 -- 40598 [ACK] Seq=1 Ack=2577 Win=89248 Len=0 TSval=3804060735 TSecr=3823488476
965	1.808895437	128.119.245.12	128.119.245.12	TLSv1.3	326	Server Hello, Change Cipher Spec, Application Data
965	1.808783361	128.119.245.12	128.119.245.12	TCP	66	40598 -- 443 [ACK] Seq=1 Ack=2577 Win=89248 Len=0 TSval=3823488557 TSecr=3804060736
967	1.808659727	128.119.245.12	128.119.245.12	TLSv1.3	148	Change Cipher Spec, Application Data
968	1.810065804	128.119.245.12	128.119.245.12	TLSv1.3	832	Application Data

845 ACK is 152953. 152953 - 1 - 152351 (content length) = 601 bytes of HTTP headers



Checkpoint 3

a. Server sent first FIN. 3 packets total to close connection

Number	Direction	Flags
2839	Server → client	FIN, ACK
2840	Client → server	FIN, ACK
2873	Server → client	ACK

b. The last data sent from the server was seq 781 and had length 582, meaning the data in the packet ran on the interval [781, 1362]. Add one for the FIN and you get 1363.

c. Server calls TIME_WAIT

capture.pcapng

ip.addr == 128.119.245.12

Packet list

Display filter

Enter a display filter ...

Find

Cancel

Options: Narrow & Wide

Case sensitive

Backwards

Multiple occurrences

No	Time	Source	Destination	Protocol	Length	Info
967	1.889639727	10.250.47.218	128.119.245.12	TLSv1.3	146	Change Cipher Spec, Application Data
968	1.818065904	10.250.47.218	128.119.245.12	TLSv1.3	532	Application Data
1017	1.884881285	128.119.245.12	10.250.47.218	TCP	66	443 → 48958 [ACK] Seq=261 Ack=3123 Win=72064 Len=0 TSval=3804060815 TSecr=3823488058
1018	1.885756657	128.119.245.12	10.250.47.218	TLSv1.3	385	Application Data
1019	1.885756987	128.119.245.12	10.250.47.218	TLSv1.3	563	Application Data
1020	1.886262576	10.250.47.218	128.119.245.12	TCP	66	40598 → 443 [ACK] Seq=3123 Ack=1877 Win=64512 Len=0 TSval=3823488034 TSecr=3804060816
2839	0.634914511	128.119.245.12	10.250.47.218	TCP	66	80 → 57792 [FIN, ACK] Seq=1363 Ack=153341 Win=249344 Len=0 TSval=3804060505 TSecr=136492532
2840	0.635243267	10.250.47.218	128.119.245.12	TCP	66	57792 → 80 [FIN, ACK] Seq=153341 Ack=1364 Win=64512 Len=0 TSval=136497534 TSecr=3804060505
2873	6.703391260	128.119.245.12	10.250.47.218	TCP	66	80 → 57792 [ACK] Seq=1364 Ack=153342 Win=249344 Len=0 TSval=3804060539 TSecr=136497534
2936	6.888330263	128.119.245.12	10.250.47.218	TLSv1.3	90	Application Data
2937	6.888330905	128.119.245.12	10.250.47.218	TCP	66	443 → 48958 [FIN, ACK] Seq=1101 Ack=3123 Win=72064 Len=0 TSval=3804060817 TSecr=3823488034
2938	6.888703844	10.250.47.218	128.119.245.12	TLSv1.3	90	Application Data
2939	6.888706923	10.250.47.218	128.119.245.12	TCP	66	48958 → 443 [FIN, ACK] Seq=3147 Ack=1102 Win=64512 Len=0 TSval=3823490867 TSecr=3804060817
2976	6.974332249	128.119.245.12	10.250.47.218	TCP	66	443 → 48958 [ACK] Seq=1102 Ack=3148 Win=72064 Len=0 TSval=3804060593 TSecr=3823493637

Frame 2839: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface wlp1s0, id 0

Ethernet II, Src: Cisco 39:70:1f (60:26:aa:39:70:1f), Dst: de:0e:d6:a6:63:8c (de:0e:d6:a6:63:8c)

Internet Protocol Version 4, Src: 128.119.245.12, Dst: 10.250.47.218

Transmission Control Protocol, Src Port: 80, Dst Port: 57792, Seq: 1363, Ack: 153341, Len: 0

Source Port: 80

Destination Port: 57792

[Stream index: 5]

[Stream Packet Number: 188]

[Conversation completeness: Complete, WITH_DATA (31)]

[TCP Segment Len: 0]

Sequence Number: 1363 (relative sequence number)

Sequence Number (raw): 878335278

[Next Sequence Number: 1364 (relative sequence number)]

Acknowledgment Number: 153341 (relative ack number)

Acknowledgment number (raw): 1388860206

1000 = Header Length: 32 bytes (8)

[Flags: 0x011 (FIN, ACK)]

0000 = Reserved: Not set

...0 = Accurate ECN: Not set

...0 = Congestion Window Reduced: Not set

...0 = ECN-Echo: Not set

....0 = Urgent: Not set

....1 = Acknowledgment: Set

....0 = Push: Not set

....0 = Reset: Not set

....0 = SYN: Not set

...1 = FIN: Set

[TCP Flags:A..F]

Window: 1948

[Calculated window size: 249344]

[Window size scaling factor: 128]

0000 de 0e d6 a6 63 8c 60 26 aa 39 70 1f 08 00 45 00 ...c'a'9p'E

0010 00 34 89 d7 40 00 24 06 1c 95 00 77 f5 0c 0a fa ...4'0's'm...

0020 2f da 89 50 a1 c9 34 5a 55 2e 4e 03 0f 2e 80 11 .../P'4Z'U.N...

0030 07 9c 56 0b 00 00 01 01 08 0a e2 bd 6f 1d 08 22 ...V.....o"

0040 05 f4

1 and 2: Closing handshake with packets from table above + First FIN expanded.



Checkpoint 4

a.

Sample	ACK Frame	SampleRTT (ms)	EstimatedRTT (ms)
1	605	86.74	86.74
2	606	86.72	86.74
3	607	85.72	86.61
4	608	85.13	86.42

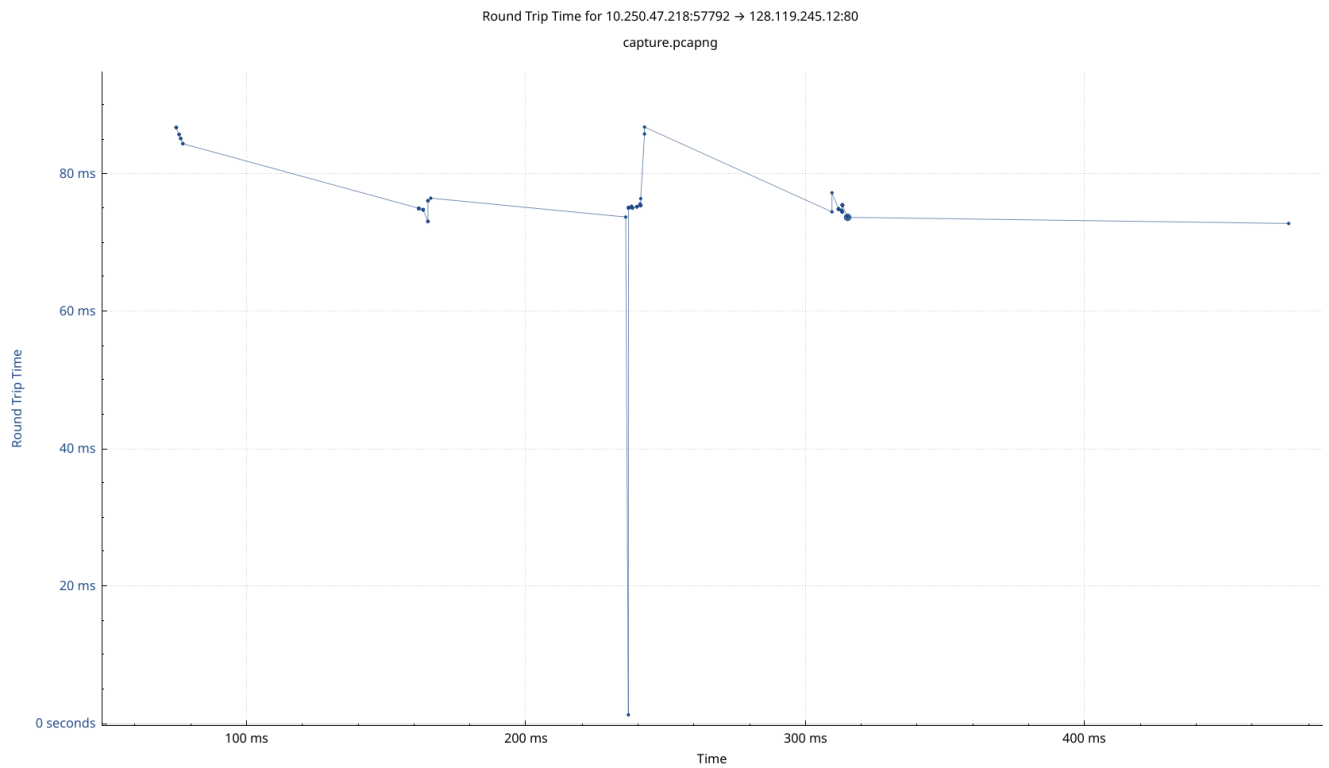
Sample 3: $0.875 \times 86.74 + 0.125 \times 85.72 = 86.61$ ms

Final EstimatedRTT 86.42 ms vs ping average of 76.71 ms vs. SYN → SYN-ACK time of 74.26 ms.

Data segments are 1374 bytes sent in bursts, so they have additional queuing delay, which causes the discrepancy.

No.	Time	Source	Destination	Protocol	Length	Info
532	0.718105471	128.119.245.12	10.250.47.218	TCP	66	443 → 52648 [ACK] Seq=26 Ack=26 Win=663 Len=0 TSval=1364909871 TSecr=2915342619
467	1.067933488	10.250.47.218	128.119.245.12	TCP	74	57792 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=136491986 TSecr=0 WS=1024
531	1.162199788	128.119.245.12	10.250.47.218	TCP	74	80 → 57792 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1386 SACK_PERM TSval=1364906093 TSecr=136491986 WS=128
532	1.162279207	10.250.47.218	128.119.245.12	TCP	66	57792 → 80 [ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval=1364902061 TSecr=3804060993
533	1.162501923	10.250.47.218	128.119.245.12	TCP	1440	57792 → 80 [ACK] Seq=1 Ack=1 Win=64512 Len=1374 TSval=1364920961 TSecr=3804060993 [TCP PDU reassembled in 776]
534	1.162501823	10.250.47.218	128.119.245.12	TCP	1440	57792 → 80 [PSH, ACK] Seq=1375 Ack=1 Win=64512 Len=1374 TSval=136492061 TSecr=3804060993 [TCP PDU reassembled in 776]
535	1.162608805	10.250.47.218	128.119.245.12	TCP	1440	57792 → 80 [ACK] Seq=2749 Ack=1 Win=64512 Len=1374 TSval=136492061 TSecr=3804060993 [TCP PDU reassembled in 776]
536	1.162612432	10.250.47.218	128.119.245.12	TCP	1440	57792 → 80 [PSH, ACK] Seq=4123 Ack=1 Win=64512 Len=1374 TSval=136492061 TSecr=3804060993 [TCP PDU reassembled in 776]
537	1.163619904	10.250.47.218	128.119.245.12	TCP	1440	57792 → 80 [ACK] Seq=5497 Ack=1 Win=64512 Len=1374 TSval=136492062 TSecr=3804060993 [TCP PDU reassembled in 776]
538	1.163617186	10.250.47.218	128.119.245.12	TCP	1440	57792 → 80 [PSH, ACK] Seq=6871 Ack=1 Win=64512 Len=1374 TSval=136492062 TSecr=3804060993 [TCP PDU reassembled in 776]
539	1.164309999	10.250.47.218	128.119.245.12	TCP	1440	57792 → 80 [ACK] Seq=8245 Ack=1 Win=64512 Len=1374 TSval=136492062 TSecr=3804060993 [TCP PDU reassembled in 776]
540	1.164248537	10.250.47.218	128.119.245.12	TCP	1440	57792 → 80 [PSH, ACK] Seq=9619 Ack=1 Win=64512 Len=1374 TSval=136492063 TSecr=3804060993 [TCP PDU reassembled in 776]
541	1.164952519	10.250.47.218	128.119.245.12	TCP	1440	57792 → 80 [ACK] Seq=10993 Ack=1 Win=64512 Len=1374 TSval=136492063 TSecr=3804060993 [TCP PDU reassembled in 776]
542	1.164954633	10.250.47.218	128.119.245.12	TCP	1440	57792 → 80 [PSH, ACK] Seq=12367 Ack=1 Win=64512 Len=1374 TSval=136492063 TSecr=3804060993 [TCP PDU reassembled in 776]
605	1.249331931	128.119.245.12	10.250.47.218	TCP	66	80 → 57792 [ACK] Seq=1 Ack=2749 Win=66832 Len=0 TSval=13649060171 TSecr=136492061
606	1.249332391	128.119.245.12	10.250.47.218	TCP	66	80 → 57792 [ACK] Seq=1 Ack=5497 Win=76136 Len=0 TSval=1364902061 TSecr=136492061
607	1.249332481	128.119.245.12	10.250.47.218	TCP	66	80 → 57792 [ACK] Seq=1 Ack=8245 Win=76896 Len=0 TSval=1364902062 TSecr=136492062
608	1.249332551	128.119.245.12	10.250.47.218	TCP	66	80 → 57792 [ACK] Seq=1 Ack=10993 Win=74132 Len=0 TSval=13649060171 TSecr=136492063
609	1.249332632	128.119.245.12	10.250.47.218	TCP	66	80 → 57792 [ACK] Seq=1 Ack=13741 Win=71424 Len=0 TSval=13649060171 TSecr=136492063
610	1.249333656	10.250.47.218	128.119.245.12	TCP	1440	57792 → 80 [ACK] Seq=13741 Ack=1 Win=64512 Len=1374 TSval=136492148 TSecr=3804060171 [TCP PDU reassembled in 776]
611	1.249401831	10.250.47.218	128.119.245.12	TCP	1440	57792 → 80 [PSH, ACK] Seq=15115 Ack=1 Win=64512 Len=1374 TSval=136492148 TSecr=3804060171 [TCP PDU reassembled in 776]
612	1.249412832	10.250.47.218	128.119.245.12	TCP	1440	57792 → 80 [ACK] Seq=16489 Ack=1 Win=64512 Len=1374 TSval=136492148 TSecr=3804060171 [TCP PDU reassembled in 776]
613	1.249414265	10.250.47.218	128.119.245.12	TCP	1440	57792 → 80 [PSH, ACK] Seq=17863 Ack=1 Win=64512 Len=1374 TSval=136492148 TSecr=3804060171 [TCP PDU reassembled in 776]
614	1.249418485	10.250.47.218	128.119.245.12	TCP	1440	57792 → 80 [ACK] Seq=19237 Ack=1 Win=64512 Len=1374 TSval=136492149 TSecr=3804060171 [TCP PDU reassembled in 776]
615	1.249418168	10.250.47.218	128.119.245.12	TCP	1440	57792 → 80 [ACK] Seq=20611 Ack=1 Win=64512 Len=1374 TSval=136492148 TSecr=3804060171 [TCP PDU reassembled in 776]
616	1.249433809	10.250.47.218	128.119.245.12	TCP	1440	57792 → 80 [PSH, ACK] Seq=21985 Ack=1 Win=64512 Len=1374 TSval=136492148 TSecr=3804060171 [TCP PDU reassembled in 776]
617	1.250990880	10.250.47.218	128.119.245.12	TCP	1440	57792 → 80 [ACK] Seq=23359 Ack=1 Win=64512 Len=1374 TSval=136492149 TSecr=3804060171 [TCP PDU reassembled in 776]
618	1.250990856	10.250.47.218	128.119.245.12	TCP	1440	57792 → 80 [ACK] Seq=24733 Ack=1 Win=64512 Len=1374 TSval=136492149 TSecr=3804060171 [TCP PDU reassembled in 776]
619	1.250991584	10.250.47.218	128.119.245.12	TCP	1440	57792 → 80 [PSH, ACK] Seq=26107 Ack=1 Win=64512 Len=1374 TSval=136492149 TSecr=3804060171 [TCP PDU reassembled in 776]
620	1.250997194	10.250.47.218	128.119.245.12	TCP	1440	57792 → 80 [ACK] Seq=27481 Ack=1 Win=64512 Len=1374 TSval=136492149 TSecr=3804060171 [TCP PDU reassembled in 776]
621	1.250999659	10.250.47.218	128.119.245.12	TCP	1440	57792 → 80 [ACK] Seq=28855 Ack=1 Win=64512 Len=1374 TSval=136492149 TSecr=3804060171 [TCP PDU reassembled in 776]
622	1.250991583	10.250.47.218	128.119.245.12	TCP	1440	57792 → 80 [PSH, ACK] Seq=30229 Ack=1 Win=64512 Len=1374 TSval=136492149 TSecr=3804060171 [TCP PDU reassembled in 776]
623	1.252679548	10.250.47.218	128.119.245.12	TCP	1440	57792 → 80 [ACK] Seq=31603 Ack=1 Win=64512 Len=1374 TSval=136492151 TSecr=3804060171 [TCP PDU reassembled in 776]
624	1.252681711	10.250.47.218	128.119.245.12	TCP	1440	57792 → 80 [ACK] Seq=32977 Ack=1 Win=64512 Len=1374 TSval=136492151 TSecr=3804060171 [TCP PDU reassembled in 776]
625	1.252684245	10.250.47.218	128.119.245.12	TCP	1440	57792 → 80 [PSH, ACK] Seq=34351 Ack=1 Win=64512 Len=1374 TSval=136492151 TSecr=3804060171 [TCP PDU reassembled in 776]
626	1.252696979	10.250.47.218	128.119.245.12	TCP	1440	57792 → 80 [ACK] Seq=35725 Ack=1 Win=64512 Len=1374 TSval=136492151 TSecr=3804060171 [TCP PDU reassembled in 776]
627	1.252699424	10.250.47.218	128.119.245.12	TCP	1440	57792 → 80 [ACK] Seq=37099 Ack=1 Win=64512 Len=1374 TSval=136492151 TSecr=3804060171 [TCP PDU reassembled in 776]
628	1.252701586	10.250.47.218	128.119.245.12	TCP	1440	57792 → 80 [PSH, ACK] Seq=38473 Ack=1 Win=64512 Len=1374 TSval=136492151 TSecr=3804060171 [TCP PDU reassembled in 776]
629	1.252697405	10.250.47.218	128.119.245.12	TCP	1440	57792 → 80 [ACK] Seq=39847 Ack=1 Win=64512 Len=1374 TSval=136492152 TSecr=3804060171 [TCP PDU reassembled in 776]
630	1.303333888	128.119.245.12	10.250.47.218	TCP	78	(TCP Dup ACK 600#1) 80 → 57792 [ACK] Seq=1 Ack=13741 Win=79816 Len=0 TSval=13649060253 TSecr=136492063 SLE=17863 SRE=19237
631	1.303430756	10.250.47.218	128.119.245.12	TCP	1440	57792 → 80 [ACK] Seq=41221 Ack=1 Win=64512 Len=1374 TSval=136492222 TSecr=3804060253 [TCP PDU reassembled in 776]
632	1.304308687	128.119.245.12	10.250.47.218	TCP	78	(TCP Dup ACK 600#2) 80 → 57792 [ACK] Seq=1 Ack=13741 Win=79816 Len=0 TSval=13649060253 TSecr=136492063 SLE=17863 SRE=19237

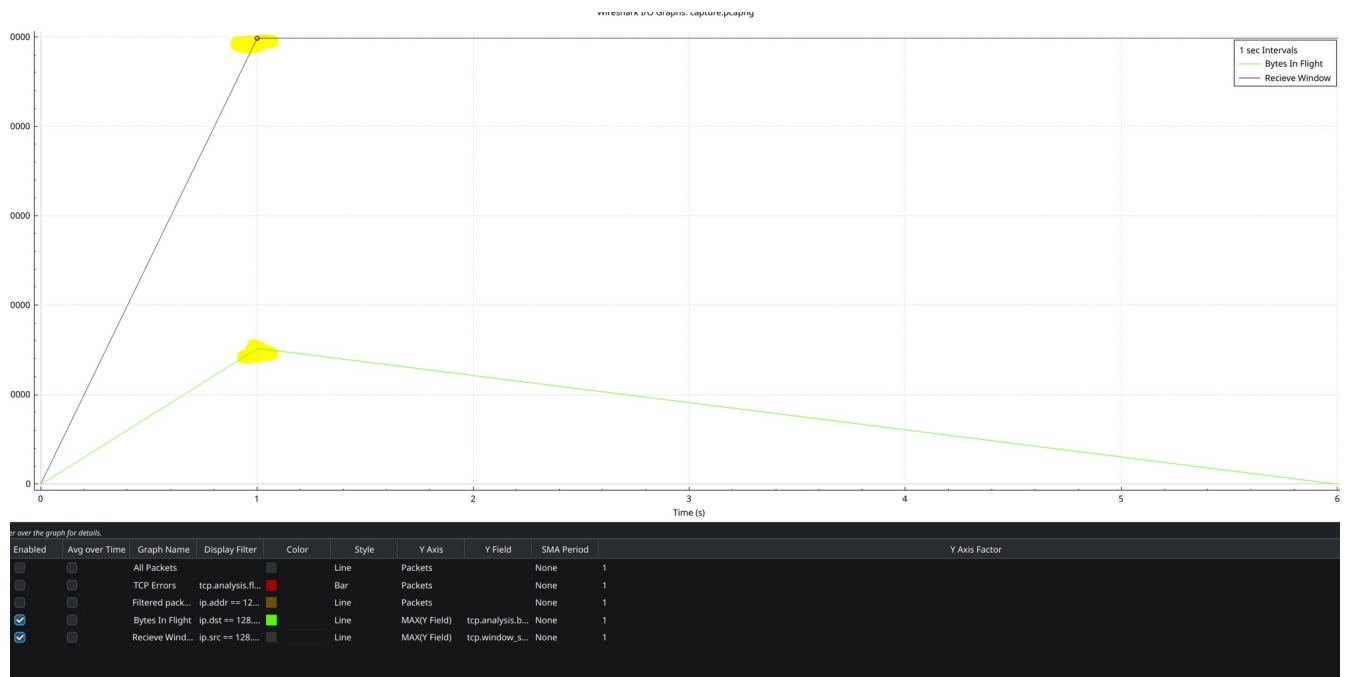
Packets 533-535 are sent from client. 605 ACKs 533-534, 606 ACKs 535



```
tpetri@clouds:~/Downloads$ ping -c 5 gaia.cs.umass.edu
PING gaia.cs.umass.edu (128.119.245.12) 56(84) bytes of data.
64 bytes from gaia.cs.umass.edu (128.119.245.12): icmp_seq=1 ttl=36 time=83.1 ms
64 bytes from gaia.cs.umass.edu (128.119.245.12): icmp_seq=2 ttl=36 time=75.2 ms
64 bytes from gaia.cs.umass.edu (128.119.245.12): icmp_seq=3 ttl=36 time=74.7 ms
64 bytes from gaia.cs.umass.edu (128.119.245.12): icmp_seq=4 ttl=36 time=73.9 ms
64 bytes from gaia.cs.umass.edu (128.119.245.12): icmp_seq=5 ttl=36 time=76.7 ms

--- gaia.cs.umass.edu ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4007ms
rtt min/avg/max/mdev = 73.861/76.713/83.064/3.312 ms
```

b. The bytes in flight never exceeds the window, meaning TCP's flow control is working. First second of steven's graph looks like a stepwise function

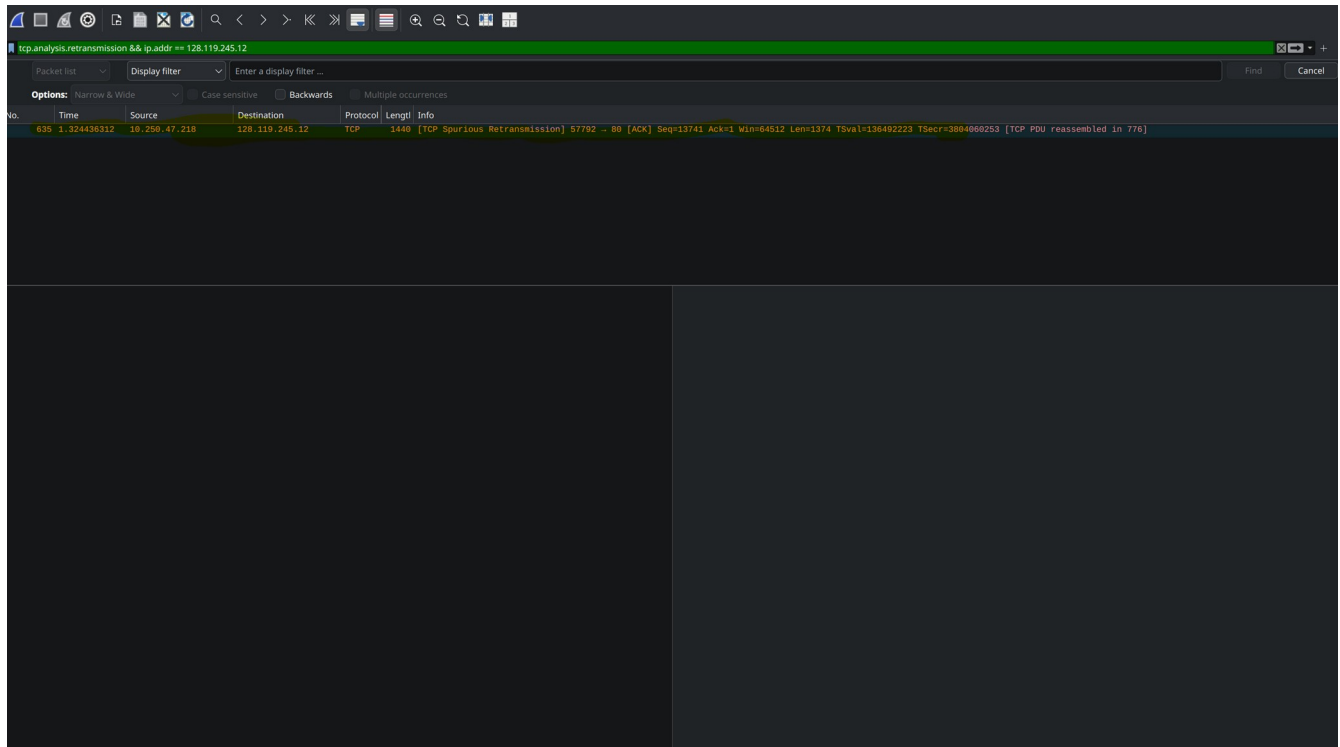


IO Graph



First Second of Stevens

c. Yes. Frame 635, seq 13741, retransmit of frame 610. Fast retransmit.



The image shows a Wireshark packet capture window with a filter set to 'tcp.analysis.retransmission && ip.addr == 128.119.245.12'. The packet list shows a single entry, packet 635, which is a TCP retransmission. The packet details pane shows the TCP header and options, including a sequence number of 13741 and a window size of 64512. The packet bytes pane is empty.

No.	Time	Source	Destination	Protocol	Length	Info
635	1.924436312	10.290.47.218	128.119.245.12	TCP	1440	[TCP Spurious Retransmission] 57792 → 80 [ACK] Seq=13741 Ack=1 Win=64512 Len=1374 TSeq=136492223 TSecr=388466253 [TCP PDU reassembled in 776]

Only one retransmission: Frame 635